

Safe but Slow: Objective Labels for Over-Conservative Driving Trajectories

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Abstract—Learned planners are trained to copy a human drive, so a hesitant drive is copied as faithfully as a decisive one. Work that adds negative examples to imitation learning has so far used unsafe trajectories, which leaves a second failure mode unaddressed: waiting when there is nothing to wait for. This paper gives a computable definition of that failure mode. A trajectory is treated as over-conservative when it is safe by the usual measures and clearly slower than the human drive, provided no traffic-engineering reason for the delay can be found. Two independent tests supply the reason. One asks whether the slower trajectory turned down a gap that drivers are known to accept, using critical gap values from the Highway Capacity Manual. The other asks whether a vehicle hidden behind an obstruction could have reached the conflict point first. On the NAVSIM navtest split the procedure labels 2,783 trajectories across 742 scenes without any human annotation. The occlusion test alone accounts for 207 of the 244 scenes it protects, so it is not redundant with the gap test. A camera-only Latent TransFuser baseline scores 94.3 on the benchmark metric in unsignalized crossings, its best stratum, yet accepts only 30% of the gaps a human accepted there, which shows the benchmark metric does not see this failure. Training with the labels is under way.

Index Terms—autonomous driving, motion planning, imitation learning, gap acceptance, occlusion

I. Introduction

A planner that never collides is not yet a planner one would want to ride behind. In merges and unprotected turns the vehicle has to weigh safety against acting in time, and worst-case reasoning tends to settle that trade-off by waiting [1], [2]. The cost is real. Hesitation delays the traffic behind and makes the vehicle’s intent hard for other drivers to read.

Imitation learning inherits this behaviour directly. The loss compares the predicted trajectory with the recorded human one, and a recorded human drive is often cautious itself. Our baseline makes the point: on the frames where the human expert progressed least, a public Latent TransFuser checkpoint reaches an ego-progress sub-score of 58.5 against the human’s 61.1. The model has copied the caution rather than corrected it.

Several recent methods do add negative examples to the training signal, but they all define a negative as something dangerous. ChauffeurNet perturbs the expert until it collides [3], and BeyondDrive generates near-expert

trajectories that violate safety and pushes the policy away from them [4]. The authors of the latter say explicitly that they do not treat low-progress trajectories as negatives. Scoring-based planners such as Hydra-MDP distil the benchmark sub-scores onto a trajectory vocabulary [5], [6], which does include a progress term, yet nothing in that supervision asks whether slowness was warranted in the first place.

The missing ingredient is a way to tell justified caution from the other kind. Slowing for a vehicle that has the right of way is correct behaviour and must not be punished. Slowing at an empty intersection is not. This paper supplies that distinction in computable form and uses it to build a labelled set of over-conservative trajectories.

The paper makes two contributions. The first is a definition of over-conservatism that runs without human annotation, built from gap-acceptance thresholds in the Highway Capacity Manual [7] and from a reachability argument about hidden vehicles [8]. A generation and filtering pipeline turns that definition into labels on NAVSIM [9], released with the per-stage attrition counts that make every threshold auditable. The second contribution is a small set of behaviour measures that expose the failure mode the benchmark score hides, reported here with the baseline numbers they produce. The training experiments that these labels were built for are running on a separate machine and appear below as work in progress.

II. Related Work

A. Negative examples in driving policies

ChauffeurNet synthesises collisions and off-road departures from perturbed expert trajectories and adds them as auxiliary losses [3]. BeyondDrive draws candidates with a flow-matching model and keeps the unsafe one closest to the expert. A repulsion term then pushes the policy away from it [4]. DriveDPO forms preference pairs from a rule-based safety score [10]. PLUTO edits the scene instead of the trajectory, removing a lead vehicle or flipping a signal [11]. Every one of these treats safety as the axis along which a trajectory can be wrong. We keep the near-expert construction that they share and change the axis to progress.

B. What the benchmarks reward

nuPlan [12] and NAVSIM [9] replaced displacement error with composite scores that weigh safety terms against a progress term. Dauner et al. showed that open-loop accuracy and closed-loop quality come apart [13], and later work found that open-loop gains often fail to transfer [14]. The progress term in these scores is normalised against a rule-based planner, which matters for us: when the rule-based planner also waits, a waiting policy loses nothing. Section V shows this happening.

C. Gap acceptance and surrogate safety

Traffic engineering has measured for decades how large a gap a driver needs. The Highway Capacity Manual tabulates critical gaps by manoeuvre, from 4.1 s for a major-road left turn to 7.1 s for a left turn from a stop-controlled approach [7]; estimation procedures go back to Raff and were refined by Troutbeck [15], [16]. Field measurements of unprotected left turns put the median accepted gap near 6.0 s, with 85% of drivers accepting 8.6 s [17]. For the safety side we use post-encroachment time, defined by Allen et al. [18] and reviewed by [19], together with the longitudinal envelope of the responsibility-sensitive safety model [20].

D. Caution under occlusion

When the sight line is blocked, slowing down is the right answer. Set-based verification places a hypothetical vehicle at the edge of the sensed region and checks whether it could reach the ego vehicle [8]; related work propagates phantom particles through unobserved lane segments [21], [22]. We reuse the deterministic special case of this idea, not to plan with it, but to decide whether a slow trajectory had an excuse.

III. When is Caution Unjustified?

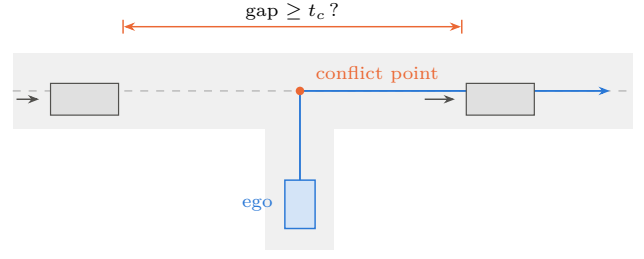
A. Setting

We work on NAVSIM [9], which replays a logged eight-second window and scores an eight-waypoint, four-second trajectory with a rule-based scorer. Writing τ^* for the human trajectory in a scene and τ for a candidate that follows the same path but moves along it differently, the question is whether τ is slower than τ^* for a reason.

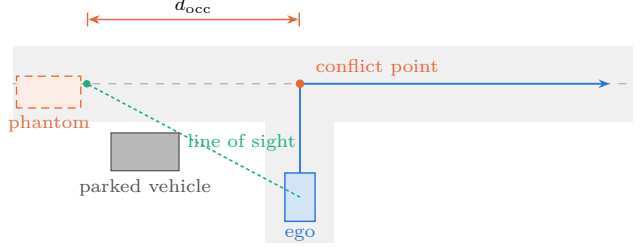
B. The definition

A candidate τ is called over-conservative when every test in Table I passes. The tests fall into two groups. The first group establishes that τ is safe and meaningfully slower, which is the easy part. The second group is the contribution: it looks for a reason to be slow and labels τ only when it finds none.

Two mechanisms can supply a reason, and they see different things. The gap test (O6) works from the observed traffic. It finds the point where the ego path crosses the path of another road user and reads off the gaps in the conflicting stream over a ten-second window. Those gaps



(a) was an acceptable gap turned down?



(b) could a vehicle one cannot see arrive first?

Fig. 1. The two ways a slow trajectory can be excused. In (a) the ego vehicle waits while a gap opens in the priority stream; the gap is compared with the critical gap t_c for that manoeuvre. In (b) no conflicting vehicle is observed at all, but a parked vehicle cuts the sight line at distance d_{occ} from the conflict point, and a phantom vehicle travelling at 1.1 times the speed limit would arrive before the ego vehicle has cleared.

are compared against the critical gap t_c for the manoeuvre at hand. A candidate that waits through a gap of at least t_c has turned down a gap that drivers accept, so its delay is unjustified. A candidate that still crosses inside the same gap, only later, is likewise adding delay for nothing, as long as its post-encroachment time stays at 1.5 s or more.

The occlusion test (O7) works from the map rather than from the observed traffic, which is what makes it complementary. It collects the lanes that cross the ego path and walks upstream along each of them. The first point the ego vehicle cannot see, because another vehicle blocks the line of sight, fixes the occluded distance d_{occ} . A phantom vehicle placed there travels at 1.1 times the posted speed limit, following [8]. If it would reach the conflict point before the decisive human drive has cleared it, plus a one-second reaction allowance, then caution was justified and the scene yields no label. This catches the case an observation-driven test cannot see, namely an intersection that looks empty because the view is blocked.

Figure 1 draws both tests on the same intersection.

C. Two kinds of caution, kept apart

It is worth separating two things that the word conservative covers. A driver can be slow while crossing, and a driver can refuse to cross at all. On navtest the human expert almost never does the second: no scene in the interaction subset shows the expert turning down a gap above the critical value. Yet the frames we call low-progress are numerous, and there the human's own

TABLE I

The tests a candidate has to pass to be labelled over-conservative. Thresholds carry a literature source; the appendix of the code repository records the sensitivity runs behind each.

Test	Threshold and source
Is the candidate safe and meaningfully slower?	
O1 Replay safety	benchmark collision and time-to-collision terms at full marks [9]
O2 Conflict-point margin	post-encroachment time ≥ 1.5 s [18], [23]
O3 Car-following margin	relative longitudinal envelope, parameters from the reference implementation of [20]
O4 Clearly slower	progress ratio to the human drive ≤ 0.8
O5 Not trivially slow	progress ratio ≥ 0.3
Is there a reason to be slow?	
O6 Gap acceptance	critical gap $t_c \in \{4.1, 6.2, 6.5, 7.1\}$ s by manoeuvre [7]
O7 Blocked sight line	phantom vehicle at $1.1 v_{lim}$, reaction allowance 1.0 s [8]

TABLE II

Scene subsets recovered by the geometric rules.

Subset	navtest	navtrain	share of navtest
Unprotected turn	802	6,057	6.6%
Merge	89	1,079	0.7%
Unsignalized crossing	205	2,671	1.7%
Union	1,093	9,738	9.0%

progress sub-score is 61.1. Human caution in this data shows up as crossing slowly. Section V shows that the planner’s caution shows up as not crossing, which is a different mechanism and needs its own measure.

IV. Building the Labels

Figure 2 shows the whole path from a raw log to a training label. Each stage is described below.

A. Finding the scenes

The OpenScene logs behind NAVSIM carry no scenario type, so the subsets are derived from map geometry. A scene is an unprotected turn when the future path runs through an unsignalized lane connector whose heading changes by at least 45 degrees. It is a merge when the lane identifier changes inside one roadblock with a vehicle nearby in the target lane. The third type, an unsignalized crossing, was added after the merge subset turned out to be small; Table II gives the sizes. Signal control is read from the map where the attribute exists and from the per-frame signal record otherwise, because the map interface leaves the turn type and the signalization flag unimplemented. Four rounds of visual sampling drove the rule revisions.

TABLE III

Generation operators and their share of the final navtest labels.

	Operator	Grid	Share
G1	Delayed start	$\Delta t \in \{1.0, 1.5, 2.0\}$ s	10.9%
G2	Scaled speed profile	$\alpha \in \{0.5, 0.65, 0.8\}$	45.7%
G3b	Unmotivated stop	decelerate and hold	42.4%
G4	Oversized gap	$\Delta \in \{1.0, 2.0\}$ s	1.0%

B. Generating slower candidates

Candidates keep the human path and change only the motion along it, so the label isolates the timing decision. Table III lists the operators. Two of them act on the speed profile, either holding the vehicle at rest for a moment or scaling the profile down. The other two manufacture a stop: one halts the vehicle where it had no reason to halt, the other waits out a conflicting vehicle the human had already passed. An earlier braking operator was dropped because its effect on four-second progress was negligible, with a median progress ratio of 0.94.

C. Filtering

Figure 3 reports what each test removes. Two observations are worth drawing out. The progress band is the coarsest filter, cutting 10,027 safe candidates to 3,699, which is by construction: most perturbations are not slow enough to matter. The occlusion test is the sharpest of the justification tests, removing 730 candidates and 221 scenes that every other test would have kept.

D. The tests do not measure the same thing

Our original protocol asked for agreement between criteria, on the assumption that a good label should be confirmed by every test. Measured agreement was near zero, with pairwise κ around 0.08. We now read that result the other way. The tests are detectors for different excuses, not repeated measurements of one construct, so low agreement is evidence of independence rather than of poor quality. Campbell and Fiske’s convergent validity applies to instruments aimed at the same construct [24], which is not our case, and the κ interpretation bands [25] do not transfer.

The useful statistic is coverage. Figure 4 shows how many scenes each test protects and how many it protects alone. The blocked sight line accounts for 244 scenes, and 207 of them are found by no other test. The replay check finds 41 of its 67 alone. Because the mechanisms barely overlap, we combine them disjunctively: any test that finds a reason cancels the label. This errs towards labelling too few, which is the safe direction for a training signal.

E. Sensitivity

The most consequential free choice is not a threshold but the rule that decides which vehicles count as conflicting traffic. Table IV tracks it. An early rule counted any vehicle passing within 2.5 m of the ego path, which swept

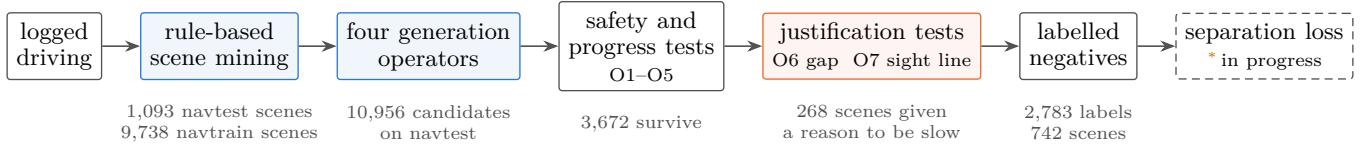


Fig. 2. From logs to labels. The two shaded stages are standard: mine the scenes where the question arises, then perturb the human trajectory into slower variants. The stage that decides whether a slow variant deserves to be a negative is the one in orange, and it is where the traffic-engineering thresholds enter. Counts are for the NAVSIM navtest split.

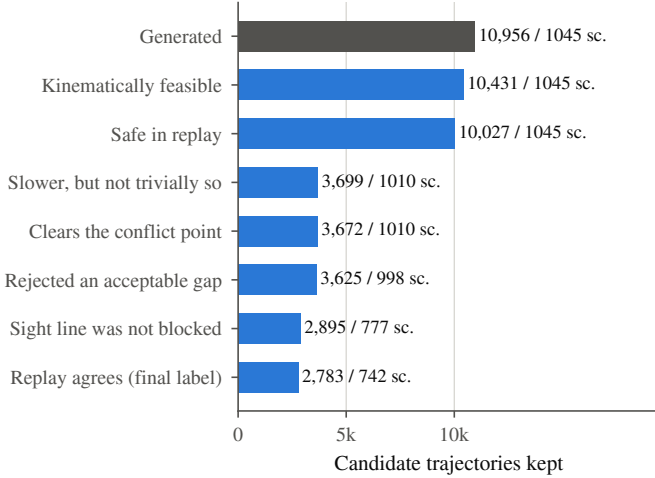


Fig. 3. Candidates surviving each test, with the number of distinct scenes they cover. The grey row is the input. Reading the two ends: of 10,956 generated candidates, 2,783 across 742 scenes become training labels.

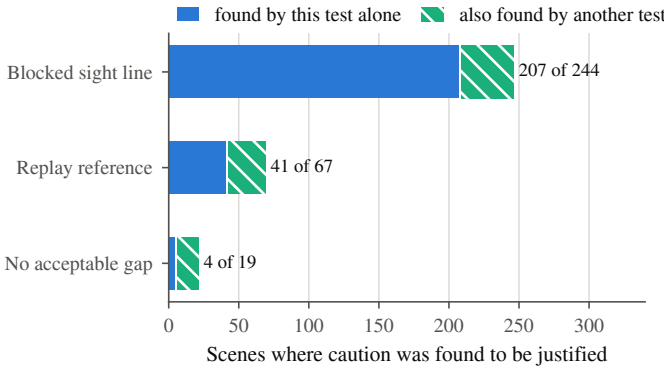


Fig. 4. Scenes in which each test finds caution to be justified, split by whether another test found the same scene. The three tests are close to disjoint, so dropping any one of them would leave that portion of the scenes mislabelled.

in queued traffic and inflated justified waiting. Excluding same-direction vehicles overcorrected, because a right turn onto a major road conflicts precisely with same-direction traffic. The rule we settled on is geometric: a vehicle counts when it starts at least 5 m away from the ego path and later comes within 2.5 m, and a vehicle directly behind the ego vehicle is treated as a follower. Merges are the known weak spot of this rule, since the conflicting vehicle there is already in the adjacent lane at the start of the window.

TABLE IV
Effect of the conflicting-traffic rule on the scene-level verdicts (navtest, 1,045 usable scenes).

Rule	With conflict	Justified wait
Any vehicle within 2.5 m	489 (47%)	224
Same-direction vehicles excluded	99 (9.5%)	35
Entering from outside the path	201 (19%)	19

V. Measuring the Behaviour

The benchmark score summarises a drive in one number, and the progress term inside it is normalised against a rule-based planner. When that planner also waits, a waiting policy is not penalised. We therefore add measures that describe what the planner did rather than how it scored, all of them decided by the same tests that produced the labels.

Two of the measures watch stopping. An unnecessary stop is a stop of at least one second that the human did not make, in a scene where no test found a reason to stop; the startup delay is the extra time the planner takes to move off relative to the human, counted only where the vehicle starts at rest. The remaining measure watches crossing. The gap acceptance rate is the share of acceptable gaps that the planner takes, restricted to gaps the human actually used, since otherwise the conflict point lies beyond the planning horizon and no trajectory could reach it. Replayed through the same code, the human drive attains the perfect value on every one of them, which is the sanity check these definitions need.

VI. Experiments

A. Setup

Labels are built on both NAVSIM splits. Evaluation uses the public Latent TransFuser checkpoints [26] with three seeds, loaded through a camera-only subclass because only the navtest camera data fits on the development machine. The full navtest score of 83.5 ± 0.45 reproduces the published 83.8, so the camera-only path is sound. Confidence intervals are from 10,000 bootstrap resamples over frames.

B. Where the baseline is weak

Table V gives the stratified scores and Figure 5 the progress term with its interval. Unprotected turns are the weakest stratum, at 79.2 against 83.5 overall, and their

TABLE V

Latent TransFuser on navtest, mean over three seeds. Sub-scores are percentages; PDMS is the composite. The rightmost column is the human drive replayed through the same scorer.

Stratum	n	NC	DAC	EP	PDMS	human
All navtest	12,146	97.7	92.3	78.3	83.5	94.8
Unprotected turn	802	99.9	86.4	71.3	79.2	93.4
Merge	89	96.8	96.3	77.4	84.1	94.5
Unsignalized crossing	205	99.8	100.0	89.3	94.3	94.1
Union of the three	1,093	99.6	89.7	75.1	82.5	93.6
Low-progress frames	198	99.7	87.2	58.5	73.9	83.8

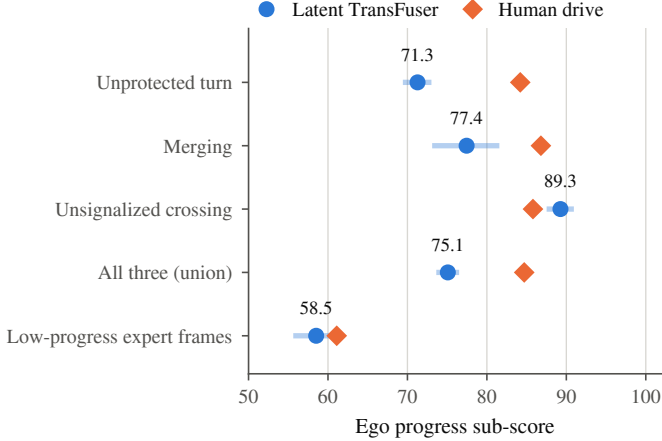


Fig. 5. Progress sub-score by stratum, with the bootstrap interval over frames. The gap to the human drive is widest in unprotected turns and closes almost completely in the low-progress stratum, where the planner has learned the human’s caution.

drivable-area term of 86.4 is the most sensitive safety sub-score in the whole table. The low-progress stratum is the one the labels target: the planner scores 58.5 there where the human scores 61.1, so it has reproduced the human’s hesitation almost exactly.

C. What the score hides

Table VI is the central empirical result of this paper. Read it next to Table V and the two disagree. Unsignalized crossings are the planner’s best stratum by composite score, at 94.3, and its best by progress, at 89.3. On gap acceptance the same stratum is the worst in the table at 30%. The explanation is the normalisation: the rule-based reference waits at unsignalized intersections too, so a waiting policy keeps a high ratio. Figure 6 puts the two views side by side. The sample behind the crossing figure is ten scenes, so we treat it as indicative and will recheck it on the larger navtrain split.

A second pattern runs through the table. Startup delay is within 0.2s of the human everywhere, so the planner is not slow to move off. What it does is decline to go: about one in five acceptable gaps is refused. This mirrors the observation in Section III that human caution takes the opposite form, and it says that a measure of hesitation built around reaction time would have found nothing here.

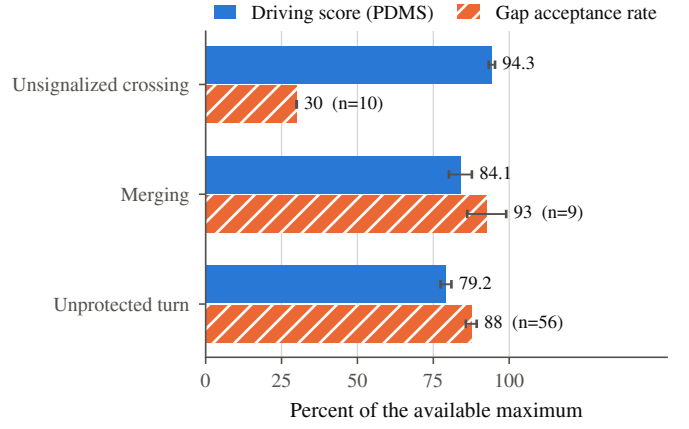


Fig. 6. The benchmark score and gap acceptance disagree. Unsignalized crossings earn the highest composite score of any stratum while the planner takes fewer than a third of the gaps a human took there. Bars show the mean over three seeds; whiskers are the bootstrap interval for the score and the seed spread for gap acceptance.

TABLE VI

Behaviour measures for the baseline, mean over three seeds with the seed spread. The human drive attains the perfect value in every column by construction.

Stratum	Unnecessary stops (%)	Startup delay (s)	Gap acceptance (% , eligible n)
Union of the three	1.53 ± 0.29	-0.01 ± 0.03	80.4 ± 0.8 (75)
Unprotected turn	0.09 ± 0.08	$+0.14 \pm 0.07$	87.5 ± 1.8 (56)
Merge	6.13 ± 0.66	$+0.05 \pm 0.01$	92.6 ± 6.4 (9)
Unsignalized crossing	5.31 ± 1.52	-0.04 ± 0.03	30.0 ± 0.0 (10)

D. Label yield

The final navtest label set holds 2,783 trajectories over 742 scenes; navtrain holds 20,799 over 5,819. Adding the occlusion test cut the navtest set by roughly a fifth and the navtrain set by a quarter, which is the price of not punishing caution that the sight line justified. Scenes where a vehicle was hidden are far more common in unsignalized crossings than in turns, at 55% against 22% on navtrain, matching the intuition that a crossing is the manoeuvre most exposed to what one cannot see.

VII. Training: Work in Progress

The labels exist to be trained on, and that step is not finished. The 8 GB development machine cannot hold the sensor data for navtrain, so training runs on a separate machine that is being provisioned. This section records the design and the acceptance criteria so that the result can be checked against a prediction rather than described after the fact.

For a regression planner the auxiliary term is a margin loss that pushes the prediction away from a labelled negative relative to the human trajectory, $L_{\text{neg}} = \max(0, m - (d(\hat{\tau}, \tau_{\text{neg}}) - d(\hat{\tau}, \tau^*)))$, with d the mean waypoint distance. The total loss adds it to the imitation loss with a weight below one, active only on frames that carry a label. A contrastive variant is implemented for

ablation. For a scoring planner the label instead marks the nearest vocabulary trajectory as a hard negative and adds a ranking term.

The acceptance criterion is stated in advance. In the interaction subset the gap acceptance rate must rise and the unnecessary-stop rate must fall, while the lower bound of the bootstrap interval on every safety sub-score stays at or above the baseline mean. In scenes where a test found caution justified, the gap acceptance rate must not rise, since a rise there would mean the model learned to be reckless rather than decisive.

Two further pieces are pending. The braking operator still yields almost nothing and needs redesigning*. Transfer to a vector-input planner in closed-loop simulation is planned after the open-loop result settles*.

VIII. Limitations

The occlusion test sees only vehicles as obstructions, because the map carries no buildings. It therefore under-states how often the view is blocked, and the bias runs towards keeping labels that a fuller model would have removed. Phantom speed comes from the posted limit, with a default where the map omits it.

The gap test degrades on merges, where the conflicting vehicle is already alongside at the start of the window and so fails the entering-from-outside rule. Safety there rests on the replay check and the car-following margin. The right measure for a merge is the headway in the target lane, which we have not implemented.

Replay is non-reactive. Other vehicles in the log have already yielded to the real, more cautious ego vehicle, so a faster reference trajectory rarely triggers a collision. This is why the replay check rejects only a small fraction on its own and why we did not rely on it alone.

Finally, the numbers in Table VI rest on small eligible samples in two strata. They are reported with their sample sizes and should be read as a direction to check, not as a settled effect.

IX. Conclusion

Waiting when there is no reason to wait is a failure mode that current training signals do not name and current benchmark scores do not see. We gave it a definition that a machine can evaluate, built from gap thresholds that traffic engineering has measured and from a reachability argument about vehicles the ego cannot see. Applied to NAVSIM the definition yields 2,783 labelled trajectories on navtest without any human annotation, and the two justification tests turn out to protect almost disjoint sets of scenes, so both are needed. Measuring the baseline against these labels showed a planner that scores best exactly where it hesitates most, which is the clearest argument we can give for not reading the composite score alone. Whether training on the labels moves the behaviour is the next question, and the criterion for answering it is fixed in advance in Section VII.

Acknowledgment

The first author thanks the members of the Nakano laboratory for discussion of the scene definitions.

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